

not an invertible matrix. By Theorem 9, $x \mapsto Ax$ is not an invertible linear transformation. $\square$

30) If A is an $n \times n$ matrix and the transformation $x \mapsto Ax$ is one-to-one, what else can you say about this transformation? Justify your answer.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements f, i and a we know that $x \mapsto Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$ and A is an invertible matrix. By Theorem 9, $x \mapsto Ax$ is invertible. $\square$

31) Suppose A is an $n \times n$ matrix with the property that the equation $Ax=b$ has at least one solution for each b in $\mathbb{R}^n$. Without using Theorems 5 or 8, explain why each equation $Ax=b$ has exactly one solution in fact.

Solution: From Theorem 4 in Chapter 1, by the equivalence of the statements (a) and (d), we know that A has a pivot position in every row. By defn, a pivot position of a matrix corresponds to the leading 1 of a row in rref. By Theorem 1 from Chapter 1 and the Row Reduction Algorithm, A can be converted to a unique rref. By condition 4 from rref, ~~each~~ each leading entry of a row must be 1. $\therefore$ There are exactly n pivots. By condition 2 from rref, for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it. $\therefore$ Each of the n columns of A has exactly 1 pivot.

By Theorem 3 from Chapter 1, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$. By defn, there are no free variables of A . By Theorem 2 from Chapter 1, the system of eqns. has a unique solution. $\square$

32) Suppose A is an $n \times n$ matrix with the property that the equation $Ax=0$ has only the trivial solution. Without using the Invertible Matrix Theorem, explain directly why the equation $Ax=b$ must have a solution for each b in $\mathbb{R}^n$.

Solution: By Theorem 3 from Chapter 1, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$. Since the solution is unique, By Theorem 2 from Chapter 1, there are no free variables in the system.